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अन्तरिक्ष प्रतिष्ठान नेपाल



Slippers2Sat-2 (S2S-2): Link Budget

Kathmandu, Nepal.



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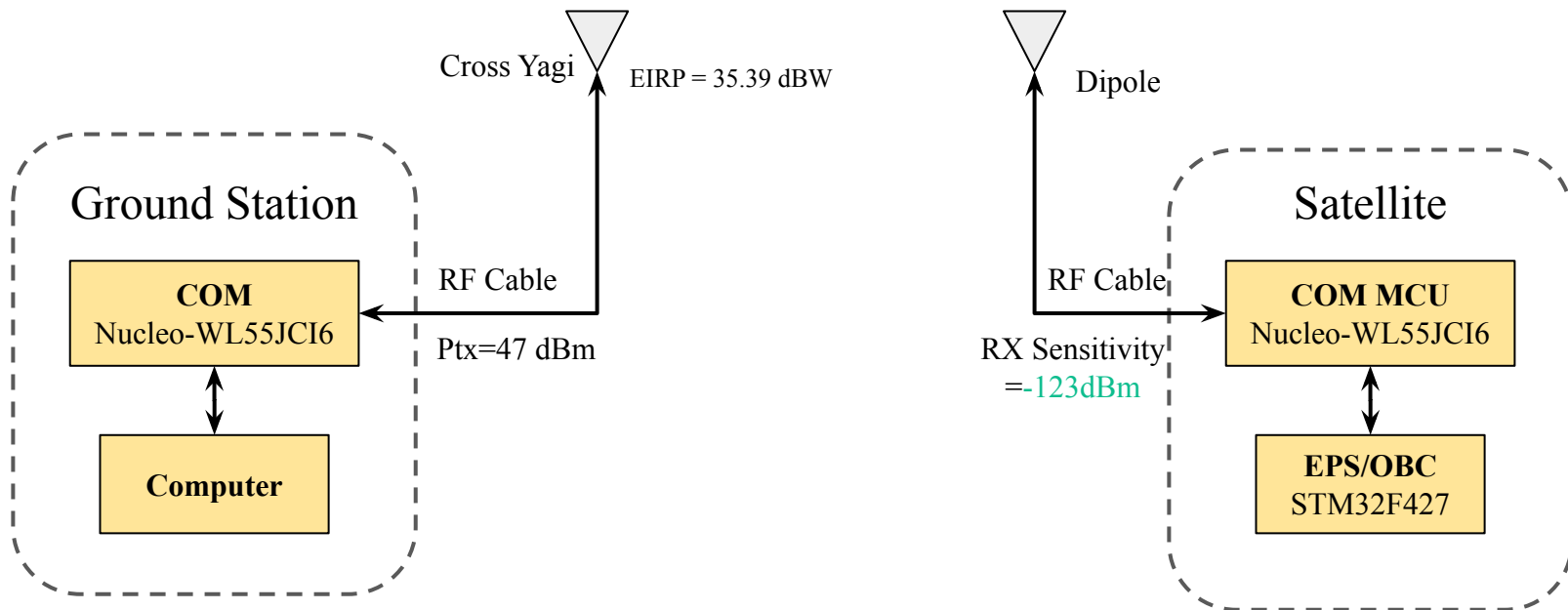
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Link Budget



- Uplink Link Margin

FSPL = 154.21 to 140.74 (EL 0 to 90)





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Link Budget



Command Uplink	
Ground Station	
TX Power [W]	50.00
TX POWER [dBW]	16.99
Line Loss [dB]	3.60
Antenna Gain [dBi]	22.00
GS EIRP [dBW]	35.39



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Link Budget



Uplink Path

Orbital Altitude [km]	600.00	600.00	600.00
Elevation Angle [degree]	0	45	90
Slant Range [km]	2829.35	814.80	600.00
GS Ant. pointing Loss [dB]	3.00	3.00	3.00
Polarization Loss [dB]	3.00	3.00	3.00
Path Loss [dB]	154.21	143.40	140.74
Atmospheric Loss [dB]	1.00	1.00	1.00
Ionospheric Loss [dB]	0.40	0.40	0.40
Isotropic Signal at Satellite [dBW]	-126.22	-115.41	-112.75



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Link Budget



Satellite Receiver			
Satellite Antenna Pointing Loss[dB]	5	5	5
Satellite Antenna Gain [dBi]	1	1	1
Line Loss [dB]	3	3	3
RX power at input [dBw]	-133.22	-122.41	-119.75
RX power at input [dBm]	-103.22	-92.41	-89.75
Receiver Sensitivity [dB]	-123	-123	-123
Link Margin [dB]	19.78	30.59	33.25



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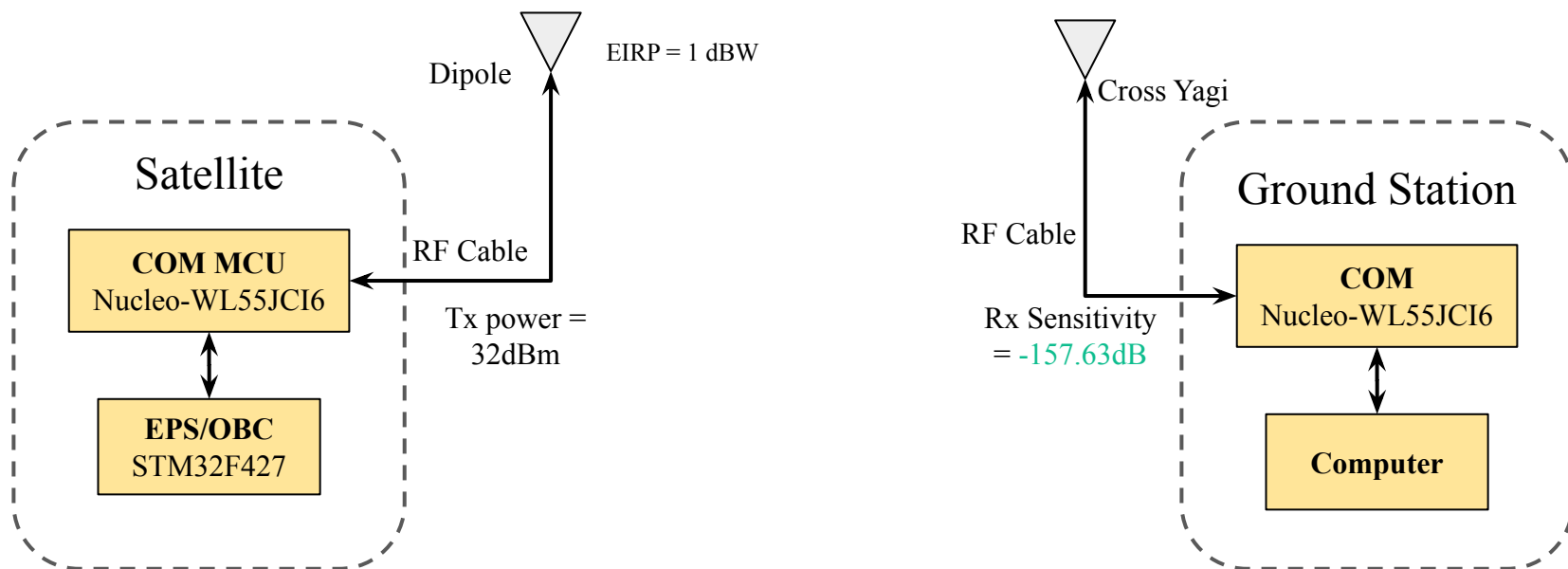
अन्तरिक्ष प्रतिष्ठान नेपाल

Link Budget



- Downlink Link Margin

FSPL = 154.25 to 140.78 (EL 0 to 90)





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Link Budget



Command Downlink	
Satellite	
TX Power [W]	1.58
TX POWER [dBW]	2.00
Line Loss [dB]	2.00
Antenna Gain [dBi]	1.00
Satellite EIRP [dBW]	1.00



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Link Budget



Downlink Path			
Orbital Altitude [km]	600.00	600.00	600.00
Elevation Angle [degree]	0	45	90
Slant Range [km]	2829.35	814.80	600.00
Satellite Ant. pointing Loss [dB]	5.00	5.00	5.00
Polarization Loss [dB]	3.00	3.00	3.00
Path Loss [dB]	154.25	143.44	140.78
Atmospheric Loss [dB]	1.00	1.00	1.00
Ionospheric Loss [dB]	0.40	0.40	0.40
Rain Loss [dB]	0.00	0.00	0.00
Isotropic Signal at GS [dBW]	-162.65	-151.84	-149.18



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Link Budget



Ground Station Receiver			
GS Antenna Pointing Loss	1	1	1
GS Antenna Gain [dBi]	22	22	22
Line Loss [dB]	3	3	3
RX power at LNA input [dBw]	-144.65	-133.84	-131.18
RX power at LNA input [dBm]	-114.65	-103.84	-101.18
Receiver Sensitivity [dB]	-157.63	-157.63	-157.63
Link Margin [dB]	42.98	53.79	56.45